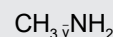
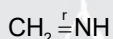
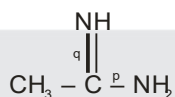
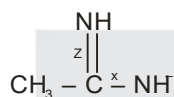
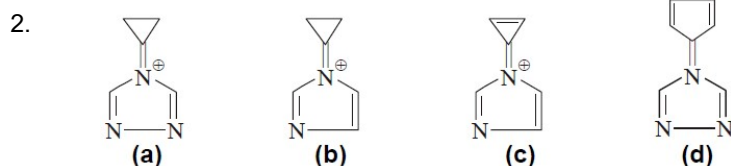


SINGLE CHOICE QUESTIONS

1. Correct order of bond length of p, q, r, x & y given in following compound is



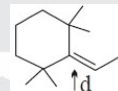
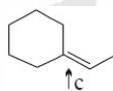
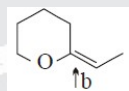
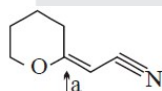
- (A) $y > x > p > q = z > r$ (B) $y > p > x = z > q > r$
 (C) $p > y > x = z > q > r$ (D) $p > y > x > q = z > r$



Correct order of basic strength of this compound is

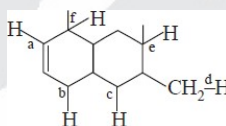
- (A) $d > b > c > a$ (B) $c > b > a > d$ (C) $c > a > b > d$ (D) $a > d > c > b$

3. Compare rotational energy barrier for the indicated bond.



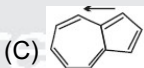
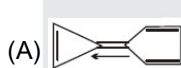
- (A) $d > c > b > a$ (B) $a > b > c > d$ (C) $c > d > a > b$ (D) $b > a > d > c$

4. Compare C-H bond energy.

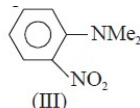
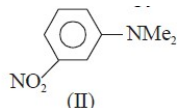
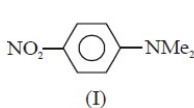


- (A) $a > b > c > d > e > f$ (B) $a > d > c > e > b > f$
 (C) $f > e > d > c > b > a$ (D) $f > b > e > c > d > a$

5. Which of the following is correct regarding direction of dipole moment?

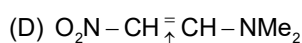
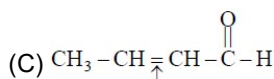
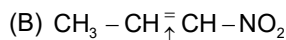
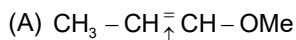


6. The correct order at resonance energy in following.

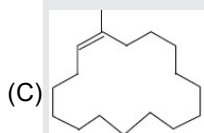
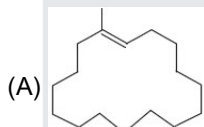


- (A) $\text{II} > \text{I} > \text{III}$ (B) $\text{III} > \text{II} > \text{I}$ (C) $\text{I} > \text{II} > \text{III}$ (D) $\text{II} > \text{III} > \text{I}$

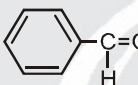
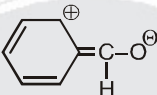
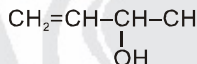
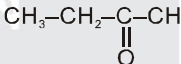
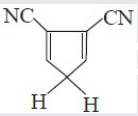
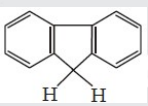
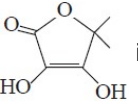
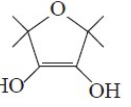
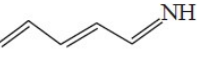
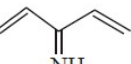
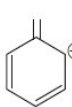
7. Which molecule has least rotational energy barrier for bond indicated by $\uparrow$.



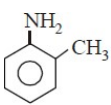
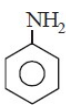
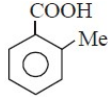
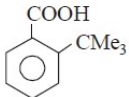
8. Statement-1 :  is more stable carbocation than .
- Statement-2 : + I effect of $\text{Me}_2\text{CH}-$ is more than that of $\text{Me}-$
- (A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1
- (B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
- (C) Statement-1 is true, statement-2 is false.
- (D) Statement-1 is false, statement-2 is true.
9. Which of the alkene will have maximum Heat of hydrogenation



MULTIPLE CHOICE QUESTIONS

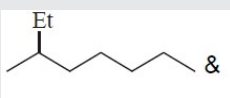
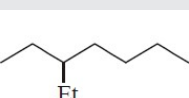
10. The tautomeric pairs are
- (A) $\text{Me}_2\text{C} = \text{NOH}$ and $\text{Me}_2\text{CH}-\text{N}=\text{O}$ (B) $\text{CH}_2=\text{CH}-\text{NHCH}_3$ and $\text{CH}_3-\text{CH}=\text{N}-\text{CH}_3$
- (C)  and  (D)  and 
11. Which of the given statement is/are not correct?
- (A)  is less acidic than $\text{CH}_2=\text{CH}-\text{CH}_2-\text{CH}=\text{CH}_2$.
- (B) $\text{CH}_3-\text{NMe}_3^+$ is more acidic than $\text{CH}_3-\text{PMe}_3^+$
- (C)  is more acidic than 
- (D)  is less acidic than 
12. Which of the following pairs has 1st having more resonance energy than 2nd.
- (A)   (B)  
- (C)   (D)  

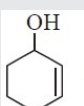
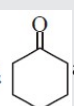
13. Select incorrect statement from the following.

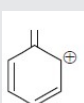
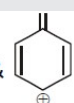
- (A)  is more basic than 
- (B)  is more acidic than 
- (C) $\text{HC} \equiv \text{CH}$ is more acidic than NH_3

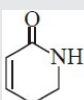
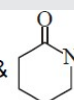
- (D)  is more stable than $\text{Me} - \text{C}(=\text{O}) - \text{O}^-$

14. Select the correct statement.

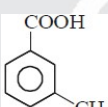
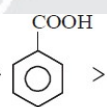
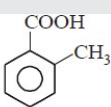
- (A)  &  are positional isomers

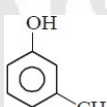
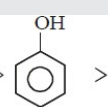
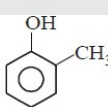
- (B)  &  are functional isomers

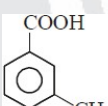
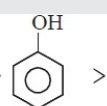
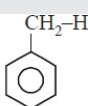
- (C)  &  have identical resonance energy.

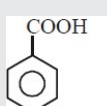
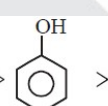
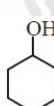
- (D)  &  have identical resonance energy.

15. Which of the following is / are correct order for acidic strength.

- (A)  >  > 

- (B)  >  > 

- (C)  >  > 

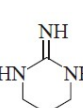
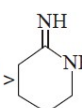
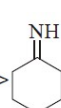
- (D)  >  > 

16. Select the correct order among the following:

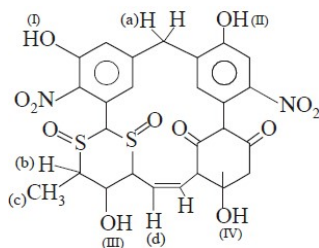
- (A) Order of heat of hydrogenation  <  < 

- (B) Order of heat of combustion  >  > 

- (C) Order of resonance energy  <  < 

- (D) Order of basic strength  >  > 

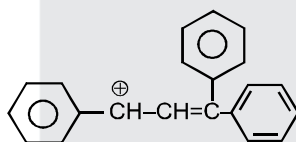
Paragraph for question nos. 17 to 19



17. The correct order of acidic strength of underlined H is
 (A) III > I > II > IV (B) I > II > IV > III (C) II > I > III > IV (D) I > II > III > IV
18. Total number of chiral centre present in the molecule is
 (A) 6 (B) 7 (C) 9 (D) 10
19. The correct order of bond energy of the marked C – H bond homolysis is:
 (A) a > b > c > d (B) d > c > b > a (C) c > b > d > a (D) d > c > a > b

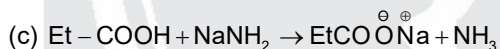
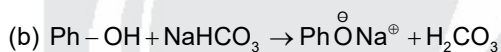
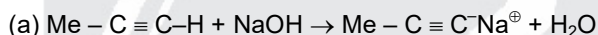
INTEGER ANSWER TYPE

20. Find the total number of the position where positive charge can be delocalized by true resonance



(Excluding the given position)

21. Fill 1 in OMR sheet if the reaction is favoured in forward direction and fill 2 in OMR sheet if the reaction is favoured in backward direction for example if all the reactions are favoured in forward direction fill 1111 as answer, and if only first two are favoured in forward direction fill 1122 in OMR sheet.



22. (a) Use the following data to answer the question below:



naphthalene

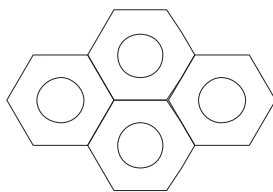
Calculate the resonance energy of naphthalene in kCal/mol.

- (b) How many grams of CO₂ is released when 84 gm of sodium bicarbonate is treated with excess of acetic acid.

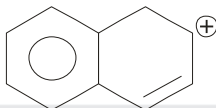
If answer of part (a) is xy and part (b) is pq then present your answer as xypq in the OMR sheet (round of your answer to nearest integers).

For example: if Answer of (a) is 12 and (b) is 13 you will fill 1213 in OMR sheet.

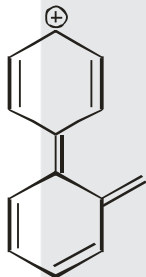
23. How many neutral (Benzoid) resonating structures are possible for.



24. Find out total no. of monoionic resonating structures of the following (Including given)

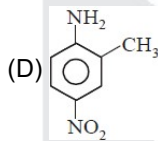
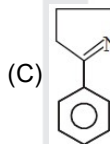
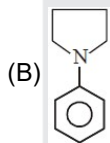


25. Find out total no. of monoionic resonating structures of the following (Including given)



26. Match the Column.

Column I



Column II

(P) Group attached to the benzene ring is +M effect showing group

(Q) Group attached to the benzene ring is -M effect showing group

(R) Group attached to the benzene ring is + H showing group.

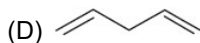
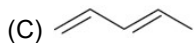
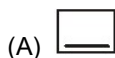
(S) Group attached to the benzene ring is -I effect showing group

(T) Compound is aromatic

- 27.

Column I

(Compounds)



Column II

(Magnitude of heat of hydrogenation)

(P) 52.5 Kcal / mole

(Q) 45.9 Kcal / mole

(R) 26.6 Kcal / mole

(S) 30.7 Kcal / mole